



Service Tool Instruction

Service Tool Instruction
Number: 3400361

Revision Level:

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2010

Description

Charge Air Cooler Leak Test Kit

Purpose

This document provides information for charge air cooler leak test kit, Part Number 4919085. The kit is used to determine if the charge air cooler (CAC) needs to be repaired or replaced for all engines.

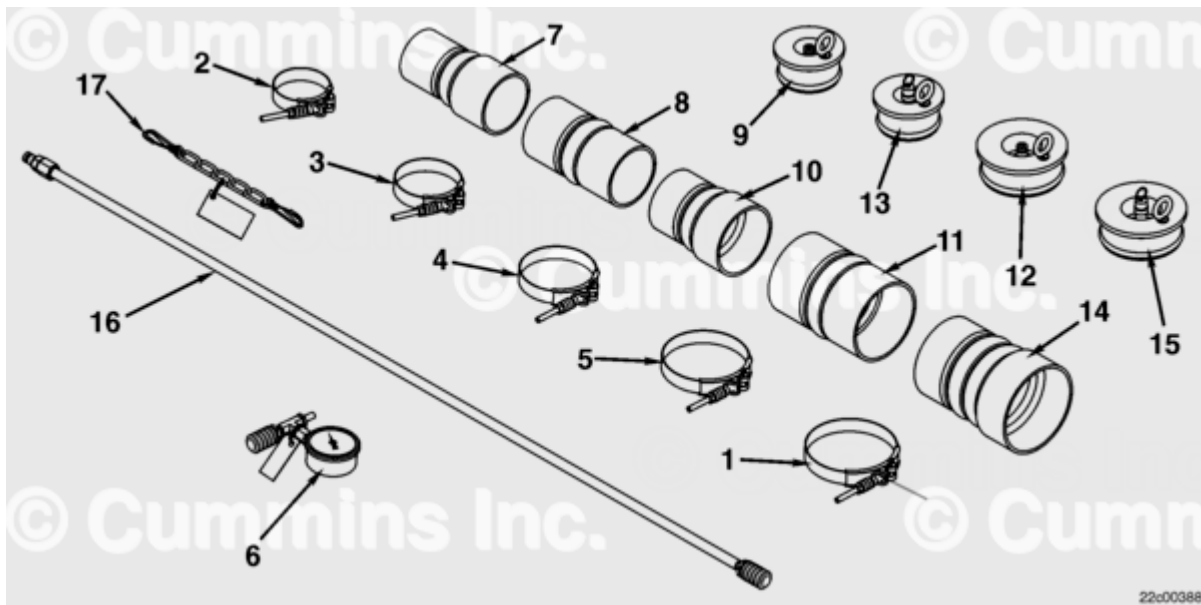


Table 1, Charge Air Cooler Leak Test Kit, Part Number 4919085

Item Number	Part Number	Description	Quantity
1	3395943	Clamp, Spring Hose	2
2	3926701	Clamp, Spring Hose	2
3	3926702	Clamp, Spring Hose	4
4	3926703	Clamp, Spring Hose	2
5	3926704	Clamp, Spring Hose	4
6	4919086	Gauge, Pressure	1
7	4919124	Hose, Flexible	2
8	4919125	Hose, Flexible	2

Table 1, Charge Air Cooler Leak Test Kit, Part Number 4919085

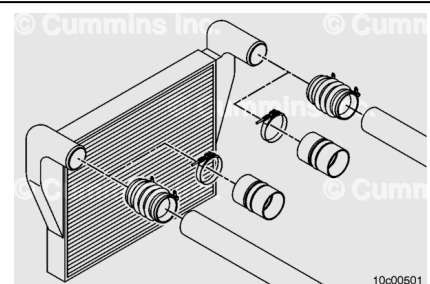
Item Number	Part Number	Description	Quantity
9	4919126	Cap, Leak Test	1
10	4919128	Hose, Flexible	2
11	4919129	Hose, Flexible	2
12	4919130	Cap, Leak Test	1
13	4919135	Cap, Leak Test	1
14	4919136	Hose, Flexible	2
15	4919137	Cap, Leak Test	1
16	4919138	Hose, Flexible	1
17	4919140	Chain, Retaining	2
Not Shown	4919739	Case, Storage	1

Disconnect all tubes and hoses from the CAC inlet and outlet.

Install the appropriate flexible hoses, included in kit, Part Number 4919085, onto the CAC.

Install the appropriate spring hose clamps included in kit, Part Number 4919085 and tighten.

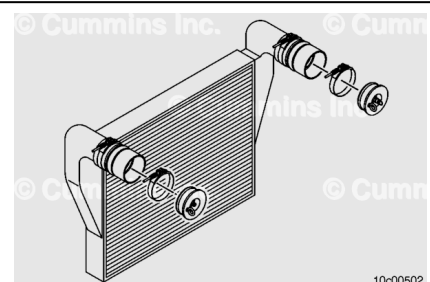
Torque Value: 8 n·m [71 in-lb]



Install the appropriate leak test caps into the hoses.

Install the appropriate spring hose clamps included in kit, Part Number 4919085 and tighten.

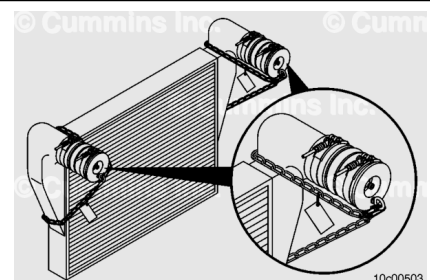
Torque Value: 8 n·m [71 in-lb]



⚠ CAUTION ⚠

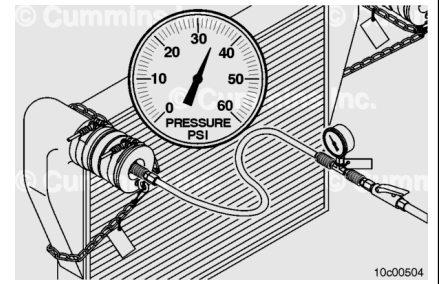
Equipment damage and personal injury can result from unrestrained leak test caps blowing out of the hoses. Be certain the plugs are properly restrained.

Install the leak test cap retaining chains in front of the CAC in such a way as to limit the movement of each cap, in case the cap blows out of the hose when air pressure is applied. The chains **must** limit the caps to the shortest possible movement.



⚠ CAUTION ⚠

Excessive air pressure can blow a leak test cap out of a hose, resulting in equipment damage or personal injury. Do NOT use more than 241 kPa [35 psi] air pressure.



Connect the flexible hose to the quick connect fitting on the leak test cap.

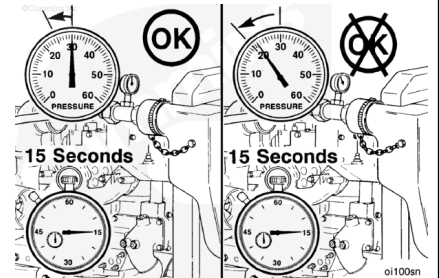
Connect the pressure gauge assembly to the flexible hose.

The safety relief valve on the leak test cap will relieve pressure at 276 kPa [40 psi]. If escaping air is heard, it is an indication that the maximum test air pressure has been exceeded.

Use shop air to pressurize the CAC while watching the pressure gauge readout. Do **not** exceed 241 kPa [35 psi].

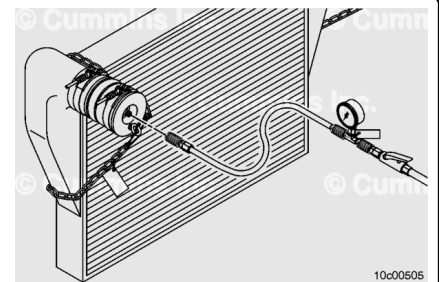
When the pressure drops to 207 kPa [30 psi], begin timing with a stop watch.

If the pressure drop is greater than 49 kPa [7 psi] in 15 seconds, the CAC **must** be replaced.



⚠ CAUTION ⚠

Once the test is completed, relieve the pressure in the CAC by disconnecting the flexible hose from the quick connect fitting on the leak test cap.



Disconnect the gauge assembly and flexible hose from the leak test cap.

Remove the leak test components from the CAC.

Note : Periodically check the relief valve on the leak test caps to ensure that they start passing air at 276 Pa [40 psi].